



# Specifying and Analyzing Transactional Consistency Models with Predicates

**Hengfeng Wei**   Si Liu   Yuxing Chen

Hunan University

Texas A&M University

Renmin University of China

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A semantic and graph-theoretic foundation for predicate operations

# Item Operations vs. Predicate Operations

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**Read**( $x, v$ )

**Write**( $x, v$ )

point access

returns one value

| Employee |        |
|----------|--------|
| EID      | Salary |
| X        | 8      |
| Y        | 10     |
| Z        | 6      |

**PredRead**( $P, M$ )

$P : \text{Salary} < 9$

condition-based access

returns a set  $M$  of **matched** values

What we know about transactional consistency models  
without predicates

# Review: Intuitive or Operational Specs

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Eswaran et al., 1976

## The Notions of Consistency and Predicate Locks in a Database System

K.P. Eswaran, J.N. Gray,  
R.A. Lorie, and I.L. Traiger  
IBM Research Laboratory  
San Jose, California

**SER**

serial execution

Berenson et al., 1995

## A Critique of ANSI SQL Isolation Levels

Hal Berenson  
Phil Bernstein  
Jim Gray  
Jim Melton  
Elizabeth O'Neil  
Patrick O'Neil

Microsoft Corp.  
Microsoft Corp.  
U.C. Berkeley  
Sybase Corp.  
UMass/Boston  
UMass/Boston

haroldb@microsoft.com  
philbe@microsoft.com  
gray@crl.com  
jim.melton@sybase.com  
coneil@cs.umb.edu  
poneil@cs.umb.edu

**SI**

snapshot + write-conflict control

# Review: Axiomatic Specs

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Cerone, Bernardi, and Gotsman, 2015

## A Framework for Transactional Consistency Models with Atomic Visibility

Andrea Cerone, Giovanni Bernardi, and Alexey Gotsman

IMDEA Software Institute, Madrid, Spain

(VIS, AR)

Crooks et al., 2017

## Seeing is Believing: A Client-Centric Specification of Database Isolation

Natacha Crooks

The University of Texas at Austin and Cornell University

Lorenzo Alvisi

The University of Texas at Austin and Cornell University

Yuer Pu

Cornell University

Allen Clement

Google, Inc.

Client-Centric Views

Complexity:OOPSLA2019

## On the Complexity of Checking Transactional Consistency

RANADEEP BISWAS, Université de Paris, IRIF, CNRS, France

CONSTANTIN ENEA, Université de Paris, IRIF, CNRS, France

Checking Complexity

# Review: Dependency Graphs

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Adya, 1999

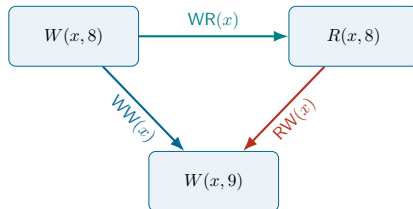
## **Weak Consistency: A Generalized Theory and Optimistic Implementations for Distributed Transactions**

by

Atul Adya

Submitted to the Department of Electrical Engineering and Computer Science  
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# Review: SER Characterization Theorem

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Adya, 1999

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Bernstein, Hadzilacos, and Goodman, 1987

**Concurrency Control  
and Recovery  
in Database Systems**

P.A. BERNSTEIN • V. HADZILACOS • N. GOODMAN

**Theorem (roughly stated)**

*An item-operation history is serializable iff  
its dependency graph is acyclic.*

# Review: SI Characterization Theorem

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Cerone and Gotsman, 2018

## Analysing Snapshot Isolation

ANDREA CERONE, Imperial College London, UK  
ALEXEY GOTSMAN, IMDEA Software Institute, Spain

### Theorem (roughly stated)

*An item-operation history satisfies SI iff every cycle of its dependency graph has at least two adjacent RW edges.*

# Review: Downstream Analysis Tasks

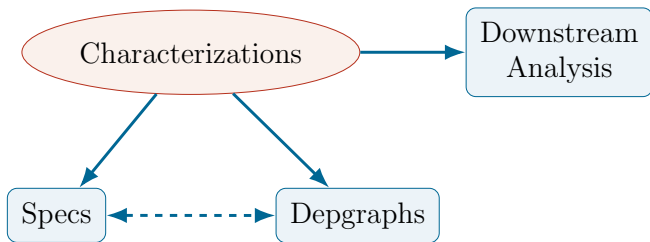
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| Task                  | Representative work                                                                |
|-----------------------|------------------------------------------------------------------------------------|
| History checking      | Kingsbury and Alvaro, 2020; Tan et al., 2020; Huang et al., 2023; Cai et al., 2025 |
| Protocol verification | Xiong, 2020                                                                        |
| Program robustness    | Bernardi and Gotsman, 2016                                                         |

## Review (Summary)

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What we know about transactional consistency models  
without predicates



What we know about transactional consistency models  
with predicates

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Eswaran et al., 1976

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Sybase Corp.  
UMass/Boston  
UMass/Boston

haroldb@microsoft.com  
philbe@microsoft.com  
gray@crl.com  
jim.melton@sybase.com  
coneil@cs.umb.edu  
poneil@cs.umb.edu

**SI**

snapshot + write-conflict control

# Review: Axiomatic Specs

Bouajjani, Enea, and Román-Calvo, 2025

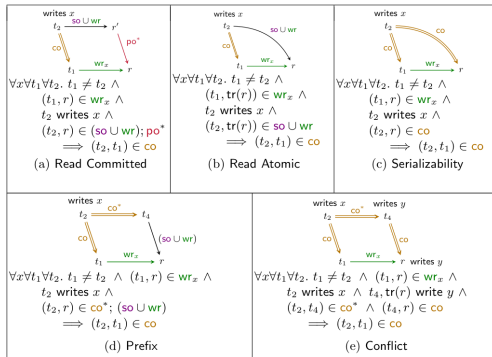
## On the Complexity of Checking Mixed Isolation Levels for SQL Transactions

Ahmed Bouajjani<sup>1</sup>, Constantin Enea<sup>2</sup>, and Enrique Román-Calvo<sup>1</sup>



<sup>1</sup> Université Paris Cité, CNRS, IRIF  
{abou, calvo}@irif.fr

<sup>2</sup> LIX, École Polytechnique, CNRS and  
Institut Polytechnique de Paris  
cenea@lix.polytechnique.fr



On checking complexity and algorithms; No dependency graph correspondence

# Review: Dependency Graphs

Adya, 1999

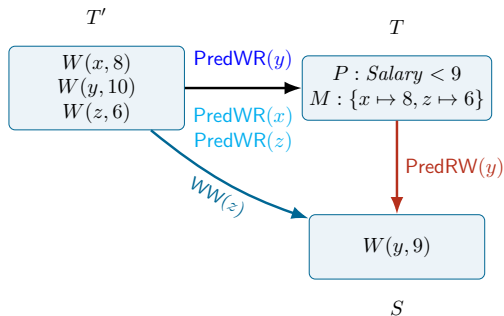
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$T$  also “observes” **unmatched**  $W(y, 10)$  of  $T'$



$S$  later than  $T'$  **changes matches** of  $T$  on  $y$

# Review: Dependency Graphs

Predicate-read-dependency,  $T \xrightarrow{\text{PredWR}(x)} S$ , definitions differ.

Predicate-anti-dependency,  $T \xrightarrow{\text{PredRW}(x)} S$ , definitions differ.

PredRW

Adya, 1999

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$S$  changes matches

PredRW<sup>+</sup>

Sun and Zou, 2025

Vbox: Efficient Black-Box Serializability Verification

Anonymous Author(s)

the “earliest” frontier  $S$

PredRW<sup>-</sup>

Complexity:CAV2025 (inspired)

On the Complexity of Checking Mixed Isolation  
Levels for SQL Transactions

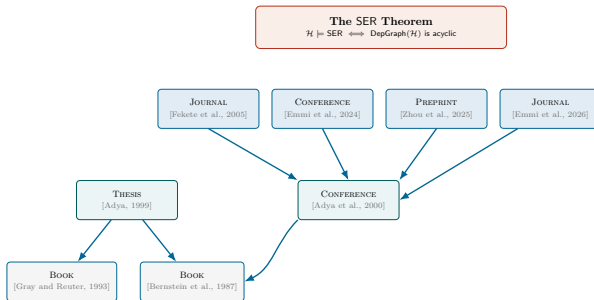
Ahmed Bouajjani<sup>1</sup>, Constantin Enea<sup>2</sup>, and Enrique Román-Calvo<sup>1</sup>



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cenea@lix.polytechnique.fr

$S$  overwrites  $x$ ,  
require no “change matches”

# Review: SER Characterization



## 3.4 Correctness and Flexibility of the New Specifications

Our conditions for PL-3 provide the well-known notion of conflict-serializability [BHG87, GR93]. That is,  $DSG(H)$  is acyclic, and G1a, and G1b are satisfied for a history  $H$  iff  $H$  is conflict-serializable. We can prove the equivalence of our PL-3 conditions with conflict-serializability using the proof given in [GR93]; our DSGs are similar to their graphs. This equivalence can also be proved along the lines of the proof given in [BHG87] for view-serializability.

No rigorous proof in the literature

# Review: SI Characterization

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**Theorem 2:** A history  $H$  consisting of committed transactions executes under Snapshot Isolation iff G1 and G-SI are disallowed.

G-SIa: Low-level *start-depends*  $\prec_t$  relation

**Definition 17:** **Snapshot Read.** All reads performed by a transaction  $T_i$  occur at its start point. That is, if  $r_i(x_j)$  occurs in history  $H$ , then:

1.  $c_j \prec_t s_i$ , and
2. if  $w_k(x_k)$  also occurs in  $H$  ( $j \neq k$ ), then either
  - (a)  $s_i \prec_t c_k$ , or
  - (b)  $c_k \prec_t s_i$  and  $x_k \ll x_j$

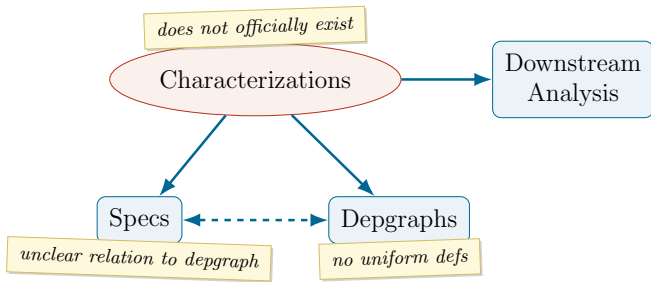
**Definition 18:** **Snapshot Write.** If  $T_i$  and  $T_j$  are concurrent, they cannot both modify the same object. That is, if  $w_i(x_i)$  and  $w_j(x_j)$  both occur in history  $H$ , then either  $c_i \prec_t s_j$  or  $c_j \prec_t s_i$ . This is the first-committer-wins property.

Proof based on *item-only* properties of SI

## Review (Summary)

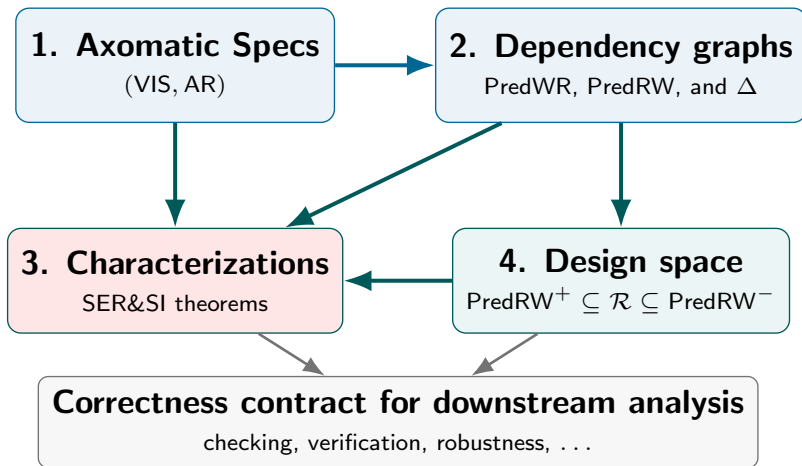
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What we know about transactional consistency models  
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# Contributions: Fill and Exploit the Gap

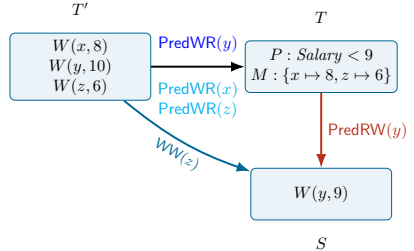
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# 1. Axiomatic Specs

Table 2: Consistency axioms, constraining an abstract execution  $(\mathcal{H}, \text{VIS}, \text{AR})$ . (Generalized based on [11])

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                     |                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| $\forall [E, \text{po}] \in \mathcal{T}, e \in E, x \in K, v \in V, F \subseteq E.$ $\left( \text{op}(e) = R(x, v) \wedge F = \{f \mid (\text{op}(f) = R(x, \_) \vee \text{op}(f) = W(x, \_)) \wedge f \xrightarrow{\text{po}} e\} \wedge F \neq \emptyset \right)$ $\implies \text{op}(\max F) = \_ (x, v)$                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                     | (INT-ITEM)                                                                                                                         |
| $\forall [E, \text{po}] \in \mathcal{T}, e \in E, f \in E, x \in K, v \in V, P \in \mathcal{P}, M \subseteq K \times V, M' \subseteq K \times V, F \subseteq E, G \subseteq E.$ $\left( \text{op}(e) = \text{PR}(P, M) \wedge \text{op}(f) = \text{PR}(P, M') \wedge f \xrightarrow{\text{po}} e \wedge \neg(\exists g \in E. \text{op}(g) = \text{PR}(P, \_) \wedge f \xrightarrow{\text{po}} g \xrightarrow{\text{po}} e) \wedge \right.$ $F = \{h \mid \text{op}(h) = W(x, \_) \wedge f \xrightarrow{\text{po}} h \xrightarrow{\text{po}} e\}$ $\implies \left( (F = \emptyset \implies ((x, v) \in M \iff (x, v) \in M')) \wedge \right.$ $\left. (F \neq \emptyset \implies ((x, v) \in M \iff (\text{op}(\max F) = W(x, v) \wedge P(x, v))) \right) \wedge$ |                                     | (INT-PRED)                                                                                                                         |
| $\left( \text{op}(e) = \text{PR}(P, M) \wedge \neg(\exists g \in E. \text{op}(g) = \text{PR}(P, \_) \wedge g \xrightarrow{\text{po}} e) \wedge \right.$ $G = \{h \mid \text{op}(h) = W(x, \_) \wedge h \xrightarrow{\text{po}} e\} \wedge G \neq \emptyset$ $\implies ((x, v) \in M \iff (\text{op}(\max G) = W(x, v) \wedge P(x, v)))$                                                                                                                                                                                                                                                                                                                                                                                                                           |                                     |                                                                                                                                    |
| $\forall T \in \mathcal{T}, x \in K, v \in V, T \vdash R(x, v) \implies \max_{\text{AR}}(\text{VIS}^{-1}(T) \cap \text{WriteTx}_x) \vdash W(x, v)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | (EXT-ITEM)                          | $\text{SO} \subseteq \text{VIS}$ (SESSION)                                                                                         |
| $\forall T \in \mathcal{T}, P \in \mathcal{P}, M \subseteq K \times V, x \in K, T \vdash \text{PR}(P, M, x) \implies$ $\boxed{x \in \text{DOM}(M) \implies \exists v \in V. (x, v) \in M \wedge \max_{\text{AR}}(\text{VIS}^{-1}(T) \cap \text{WriteTx}_x) \vdash W(x, v)} \wedge$ $\boxed{x \in K \setminus \text{DOM}(M) \implies \forall v \in V. (\max_{\text{AR}}(\text{VIS}^{-1}(T) \cap \text{WriteTx}_x) \vdash W(x, v) \implies \neg P(x, v))}$                                                                                                                                                                                                                                                                                                          |                                     | (EXT-PRED)                                                                                                                         |
| $\text{AR} ; \text{VIS} \subseteq \text{VIS}$ (PREFIX)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\text{VIS} = \text{AR}$ (TOTALVIS) | $\forall S, T \in \mathcal{H}. S \models T \implies (S \xrightarrow{\text{VIS}} T \vee T \xrightarrow{\text{VIS}} S)$ (NOCONFLICT) |



## External predicate consistency (EXT-PRED axiom)

For each externally observed key  $x$ , choose the **AR-latest** writer visible to the reader. Its value must equal the returned value when  $x$  **appears** in the result, and must fail the predicate when  $x$  is **omitted**.

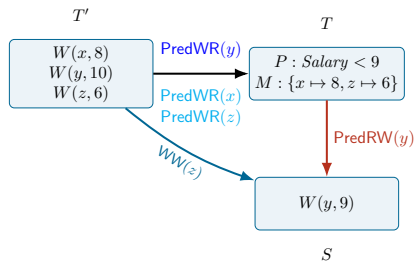
## 2. Dependency Graphs

### Graph witness

$T' \xrightarrow{\text{PredWR}(y)} T$ :  $T'$  supplies the version by which  $T$  externally observes  $x$ .

### Observation-changing successor

$T \xrightarrow{\text{PredRW}(y)} S$ :  $T' \xrightarrow{\text{WW}(y)} S$  and  $S$   
**changes the matches** of  $T$  on  $y$  wrt  $T'$ .



### 3. Characterizations

#### Theorem (SER Characterization)

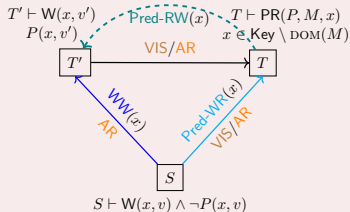
$\mathcal{H} \models \text{SER}$  iff it admits an *acyclic* predicate-aware depgraph.

#### Theorem (SI Characterization)

$\mathcal{H} \models \text{SI}$  iff it admits a predicate-aware depgraph where every cycle has two adjacent RW/PredRW edges.

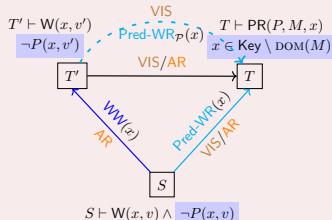
“if” direction: construct legal execution  $\mathcal{A} \models \text{Ext-Pred}$  from graph

#### Challenge: mismatched witnesses



violation of  $\text{EXT-PRED} \Rightarrow \text{PredRW} \Rightarrow$  forbidden cycle

#### Challenge (SI): mismatched witnesses



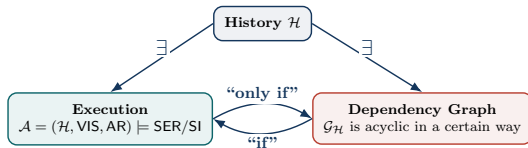
no need to recover the exact graph from execution  $\mathcal{A}$

## 4. Design Space (of Predicate Anti-Dependencies)

$$\text{PredRW}^+ \subseteq \text{PredRW} \subseteq \text{PredRW}^-$$

$$\text{PredRW}^+ \subseteq \mathcal{R} \subseteq \text{PredRW}^-$$

Every relation  $\mathcal{R}$  preserves the same existential SER/SI characterization.



"if":  $\text{PredRW}^+ \subseteq \mathcal{R}$

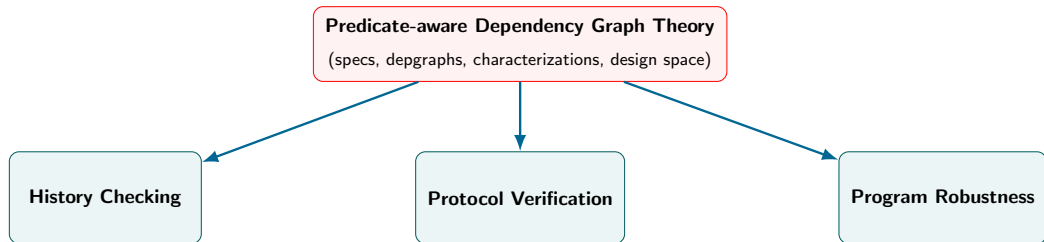
$\text{PredRW}^+ + \text{WW}$  recovers  $\text{PredRW}$  in reachability

"only if":  $\mathcal{R} \subseteq \text{PredRW}^-$

$\text{PredWR} + \text{WW}$  suffices to derive safe AR orientation

# Future Work: Downstream Analysis

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## $[\text{PredRW}^-, \text{PredRW}^+]$ -guided history checker

- a lower persistent cycle can reject
- an upper acyclic graph can accept
- ambiguous cases are refined or solved

## Takeaways: Theory Established, Downstream Analysis Enabled

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1. **Specs.** Extend (VIS, AR) with item-wise external predicate observations.
2. **Dependency graphs.** Formally define PredRW with the “change matches” condition.
3. **SER/SI Characterizations.** Prove the same execution–graph equivalence style as the item-only results, while resolving the predicate-specific SI witness ambiguity.
4. **Proof-theoretic design space.** Every relation  $\text{PredRW}^- \subseteq \mathcal{R} \subseteq \text{PredRW}^+$  preserves the same existential history-level characterization.

Checking, Verification, Robustness, ...